MILITARY SPECIFICATION
CATAPULT, AIRCRAFT EJECTION SEAT, M4A1
METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers metal parts assembly for Catapult, Aircraft Ejection Seat, M4A1. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military
- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-413 - Visual Inspection Guide for Rubber O-rings
- MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Munitions Command
- F8593670 - Catapult, Aircraft Ejection Seat, M4A1, Metal Parts Assembly
- F8797383 - Ammunition Container, Fiber, M301 for Catapult, Aircraft Ejection Seat, M4A1 or M4
- F8797384 - Box, Packing, Ammunition for Catapult, Aircraft Ejection Seat, M4A1 or M4 in Fiber Container M301
- IEL-8593670 - Index to Inspection Equipment List for Catapult, Aircraft Ejection Seat M4A1, Metal Parts Assembly

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties.- Physical properties required by the drawing shall be uniform throughout each part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure.- The outside, inside and telescoping tube shall withstand a constant internal pressure of 3000 pounds per square inch (psi) minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

3.6 Shear pin strength.- After application of protective finish, the pin shall shear when a load of 400 $\pm$ 50 pounds is applied at the steady rate of .25 inches per minute.

3.7 Structural strength.- The metal parts assembly shall be capable of withstanding each test load for 15 seconds, minimum, as indicated in Table I. The maximum allowable change between mounting points, resulting from each load application shall not exceed the values indicated in Table I. A preload of 500 pounds shall be utilized prior to each test load application.

TABLE I

Test Loads (min)	Change Between Mounting Points (max)
1. 8000 pounds in tension	.005 inches
2. 8000 pounds in tension	.001 inches
3. 2000 pounds in compression	.005 inches
4. 2000 pounds in compression	.001 inches

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:30Z
Check the source to verify that this is the current version before use.

3.8 Torque.- Torque requirements shall be as specified in Table II. No loosening or deformation shall occur upon application of the breakaway torque.

3.8.1 Bottoming torque.- The bottoming torque between block and inside tube and between ring stop and ring tube stop shall not exceed the running torque by more than 50 inch pounds.

TABLE II

Component	A Run-On (inch pounds)	Seating or Bottoming (including A)(inch pounds)	C Breakaway (inch pounds)
Spacer and Telescoping Tube			200 min.
Cap and Outside Tube	75 min.	500 + 50	400 min.
Ring, Tube Stop and Outside Tube	75 min.	500 + 50	400 min.
Ring, Stop and Ring, Tube Stop	75 min.		
Block and Inside Tube	75 min.		

3.9 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.10 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.11 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal defects. The component shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.11.2 Burr. The component shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 Foreign matter. The component or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

4.1 (Cont.)

acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Initial production sample.- At the beginning of regular production the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies and 10 sets of parts and 50 shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- The lot of metal parts assemblies shall consist of 614 catapults maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic) the criteria in Table III shall be used. For total defects within a major or minor group the

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:30Z
Check the source to verify that this is the currentversion before use.

4.3.2 (Cont.)

criteria in Table IV shall be used. All nonconforming material shall be rejected. Inspection for rubber o-rings shall be in accordance with MIL-STD-413. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE IV

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects	110	150	225	300
No. of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over - 43	6	8	11	14
Sample Sizes for Minor Defects	**110**	**150**	**225**	**300**
No. of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over - 48	8	11	17	20

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

4.3.2.1 Latch, Drawing C8593645, covering a detail of Drawing F8593670.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter of firing pin relief	Gage
102.	Length to inside surface, small end	Gage
103.	Length of cutout	Gage
104.	Width at small end	Gage
105.	Width at cutout	Gage
106.	Thickness at datum locations	Gage
107.	Location of firing pin relief	Gage
108.	Angle at large end	Gage
109.	Metal defective	Visual
110.	Protective finish inadequate or defective	Visual
111.	Surface finish of base chamfer and face improper	Visual
112.	Burr or sharp corner	Visual
113.	Foreign matter	Visual
MINOR		
201.	Total length, max.	Gage
202.	Thickness, max.	Gage
203.	Thickness at firing pin relief, max.	Gage
204.	Width at large end	Gage

4.3.2.2 Pin, Shear, Dwg. B8593654, covering a detail of Dwg. F8593670.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Diameter, max.	Gage
102.	Total length, min.	Gage
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
201.	Total length, max.	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection

4.3.2.3 Pin Firing, Dwg. C8593657 covering a detail of Dwg. F8593670.

CRITICAL

1.	Diameter of striker, max.	Gage

MAJOR

101.	Concentricity of striker with striker shoulder	Gage
102.	Concentricity of flange with striker shoulder	Gage
103.	Concentricity of groove with flange	Gage
104.	Symmetry of forward datum location with striker shoulder	Gage
105.	Symmetry of width of body with striker shoulder	Gage
106.	Symmetry of thickness of body with striker shoulder	Gage
107.	True position of pin hole with striker shoulder and thickness of body	Gage
108.	Diameter of striker shoulder, min.	Gage
109.	Diameter of flange, min.	Gage
110.	Diameter of groove	Gage
111.	Depth of hole in striker shoulder (max)	Gage
112.	Length of striker, min.	Gage
113.	Width of body, max.	Gage
114.	Width of groove	Measure
115.	Relief groove on striker shoulder, missing	Visual
116.	Metal defective	Visual
117.	Protective finish, inadequate or defective	Visual
118.	Surface finish of body, flange and striker shoulder, improper	Visual
119.	Burr or sharp corner, except as permitted	Visual
120.	Foreign matter	Visual

MINOR

201.	Diameter of striker shoulder, max.	Gage
202.	Diameter of flanges, max.	Gage
203.	Diameter of hole in striker shoulder	Gage
204.	Depth of hole in striker shoulder, min.	Gage
205.	Width at forward datum location	Gage
206.	Width at rear datum location	Gage
207.	Location of hole	Gage
208.	Surface variation on rear face of striker shoulder	Gage

4.3.2.4 Seal, Plain, Dwg. B8593658, covering a detail of Dwg. F8593670.

CRITICAL

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MAJOR	
101. Diameter of wire, max. (after plating)	Gage
102. Number of coils incorrect	Visual
103. Location variance of ends, max.	Visual
104. Metal defective	Visual
105. Protective finish, inadequate or defective	Visual
106. Burr or sharp edge	Visual
107. Foreign matter	Visual
MINOR	
201. Diameter, outside	Measure
202. Direction of turn, incorrect	Visual

4.3.2.5 Seal, Fixed, Dwg. B8593661 covering a detail of Dwg. F8593670.

CRITICAL	
None defined	
MAJOR	
101. Diameter of wire, max. (after plating)	Gage
102. Length of tang	Gage
103. Location variance of ends, max.	Visual
104. Number of coils, incorrect	Visual
105. Radius of tang band, max.	Visual
106. Metal defective	Visual
107. Protective finish inadequate or defective	Visual
108. Burr or sharp edge	Visual
109. Foreign matter	Visual
MINOR	
201. Diameter, outside	Gage
202. Direction of turn incorrect	Visual

4.3.2.6 Block, Dwg. F8593672, covering a detail of Dwg. F8593670.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Pitch diameter of pipe thread	Gage
104. Minor diameter of pipe thread, max.	Gage
105. Concentricity of counterbore with small cavity	Gage

Source: https://assist.dla.mil -- Downloaded: 20261O-22T06:30Z
Check the source to verify that this is the current version before use.

	Defect	Method of Inspection
MAJOR (Cont)		
106.	Concentricity of pitch diameter of thread with counterbore	Gage
107.	Concentricity of large cavity with counterbore	Gage
108.	Concentricity of flange with body	Gage
109.	Symmetry of length of slot with body	Gage
110.	Symmetry of thickness of rectangular section with body	Gage
111.	Symmetry of width of rectangular section with body	Gage
112.	Perpendicularity of hole with rectangular section	Gage
113.	True position of pin hole with small cavity	Gage
114.	True position of hole in rectangular section with body	Gage
115.	True position of screw hole with counterbore and base of large cavity	Gage
116.	Diameter of mounting hole in rectangular section	Gage
117.	Diameter of small cavity	Gage
118.	Diameter of counterbore, max.	Gage
119.	Depth to bottom of small cavity (max.)	Gage
120.	Length to flange, max.	Gage
121.	Length of effective thread, min.	Gage
122.	Width of rectangular section, max.	Gage
123.	Diameter of thread undercut, max., if applicable	Measure
124.	Depth to bottom of drilled hole, max.	Gage
125.	Location of pipe hole, max.	Measure
126.	Thread damaged or not full	Visual
127.	Metal defective	Visual
128.	Protective finish inadequate or defective	Visual
129.	Surface finish of small cavity, and body slots, improper	Visual
130.	Burr	Visual
131.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, min.	Gage
202.	Diameter of large cavity	Gage
203.	Diameter of counterbore, min.	Gage
204.	Diameter of flange, max.	Gage
205.	Diameter of pin hole in body	Gage
206.	Diameter of body (max.)	Gage
207.	Diameter of screw hole, min.	Gage
208.	Depth of large cavity	Gage
209.	Length from flange to base of large cavity	Gage
210.	Length of slot in body	Gage
211.	Width of slot in body	Gage
212.	Location of slot to base of large cavity	Gage
213.	Location of screw hole	Gage
214.	Location of hole in rectangular section	Gage
215.	Location of pin hole in body	Gage
216.	Depth of counterbore	Measure
217.	Profile of end of rectangular section	Measure

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:30Z
Check the source to verify that this is the current version before use.

4.3.2.7 Ring, Tube, Stop, Dwg. D8593673-74 covering a detail of Dwg. F8593670.

	Defect	Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Pitch diameter of internal thread, max.	Gage
102.	Minor diameter of internal thread, max.	Gage
103.	Pitch diameter of external thread, min.	Gage
104.	Major diameter of external thread, min.	Gage
105.	Concentricity of pitch diameter of internal thread with cavity	Gage
106.	Concentricity of pitch diameter of external thread with cavity	Gage
107.	Diameter of cavity	Gage
108.	Depth of thread cavity, min.	Gage
109.	Depth of spiral groove, max.	Gage
110.	Depth of pellet hole (max)	Gage
111.	Thread damaged or not full	Visual
112.	Pellet missing	Visual
113.	Metal defective	Visual
114.	Protective finish inadequate or defective	Visual
115.	Surface finish of cavity, improper	Visual
116.	Burr	Visual
117.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of internal thread, min.	Gage
202.	Pitch diameter of external thread, max.	Gage
203.	Length of effective internal thread, min.	Gage
204.	Depth of spiral groove, min.	Gage
205.	Depth of spiral groove relief	Gage
206.	Width of spiral groove relief	Gage
207.	Diameter of counterbore	Measure
208.	Diameter of hole in spiral groove	Measure
209.	Depth of counterbore	Measure
210.	Length of spiral groove relief	Measure
211.	Location of spiral groove relief	Measure
212.	Location of hole in spiral groove	Measure
213.	Symmetry of hole with profile of spiral groove	Visual
214.	Profile of spiral groove	Visual
215.	Profile of spiral groove relief	Visual
216.	Number of turns of spiral groove, incorrect	Visual
217.	Direction of turn of spiral groove, incorrect	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:30Z
Check the source to verify that this is the current version before use.

4.3.2.8 Ring, Stop, Drawing C8593675-76 covering a detail of Drawing F8593670.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of bore with pitch diameter of thread	Gage
104. Concentricity of counterbore with bore	Gage
105. Symmetry of width across flats with flange	Gage
106. Diameter of bore, max.	Gage
107. Diameter of chamfer at datum location	Gage
108. Diameter of thread undercut, min.	Gage
109. Length to flange	Gage
110. Length to counterbore	Gage
111. Depth of pellet hole (max)	Gage
112. Thread damaged or not full	Visual
113. Pellet missing	Visual
114. Metal defective	Visual
115. Protective finish inadequate or defective	Visual
116. Surface finish of angle of bore and bore, improper	Measure/Visual
117. Burr or sharp corner	Visual
118. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of counterbore	Gage
203. Diameter of flange	Gage
204. Diameter of bore, min.	Gage
205. Total length	Gage

4.3.2.9 Spacer, Dwg. C8593678 covering a detail of Dwg. F8593670.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of flange with pitch diameter of thread	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MAJOR (Cont.)	
104. Holes in flange missing	Visual
105. Diameter of flange, min.	Gage
106. Thread damaged or not full	Visual
107. Metal defective	Visual
108. Protective finish inadequate or defective	Visual
109. Surface finish on flange diameter, improper	Visual
110. Burr	Visual
111. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, min.	Gage
202. Diameter of flange, max.	Gage
203. Diameter of body, max.	Gage
204. Diameter of holes	Gage

4.3.2.10 Plug, Dwg. C8593679, covering a detail of Dwg. F8593670.

CRITICAL	
1. Diameter of boss hole, min.	Gage
MAJOR	
101. Concentricity of boss hole with cavity	Gage
102. Concentricity of flange with cavity	Gage
103. Concentricity of body with flange	Gage
104. Symmetry of base slot with a cavity	Gage
105. True position of pitch diameter of set screw thread with flange including perpendicularity with slot	Gage
106. True position of pin hole with cavity and slot	Gage
107. Diameter of cavity	Gage
108. Diameter of flange, min.	Gage
109. Depth of cavity, min.	Gage
110. Depth of slot	Gage
111. Total length	Gage
112. Height of boss	Gage
113. Diameter of boss (max)	Measure
114. Metal defective	Visual
115. Protective finish inadequate or defective	Visual
116. Surface finish of cavity and slot, improper	Visual
117. Burr	Visual
118. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread	Gage
202. Minor diameter of thread, max.	Gage
203. Diameter of flange, max.	Gage
204. Diameter of body	Gage
205. Diameter of pin hole, min.	Gage
206. Length of effective thread, min.	Gage
207. Width of slot	Gage
208. Location of pin hole	Gage
209. Location of threaded hole	Gage

4.3.2.11 Tube, Telescoping, Dwg. D8593680-81 covering a detail of Dwg. F8593670.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with datum body location	Gage
104. Diameter of undercut, min.	Gage
105. Diameter of body	Gage
106. Diameter of shoulder	Gage
107. Diameter, inside	Gage
108. Total length, min.	Gage
109. Thickness of wall variation, max.	Gage
110. Straightness	Gage
111. Pellet missing	Visual
112. Depth of pellet hole (max)	Gage
113. Diameter of holes in shoulder	Measure
114. Number of holes in shoulder, incorrect	Visual
115. Metal defective	Visual
116. Protective finish inadequate or defective	Visual
117. Burr	Visual
118. Foreign matter	Visual
MINOR	
201. Total length (max)	Gage

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:30Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection

4.3.2.12 Tube, Inside Dwg. D8593682-83 covering a detail of Dwg. F8593670.

CRITICAL

None defined

MAJOR

101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of pitch diameter of thread with shoulder and bore at datum locations	Gage
104.	Diameter of shoulders	Gage
105.	Thickness of wall, min.	Gage
106.	Length of effective thread, min.	Gage
107.	Straightness	Gage
108.	Depth of pellet hole, max.	Gage
109.	Pellet missing	Visual
110.	Metal defective	Visual
111.	Thread damaged or not full	Visual
112.	Protective finish inadequate or defective	Visual
113.	Surface finish of body at ends, improper	Visual
114.	Burr	Visual
115.	Foreign matter	Visual

MINOR

201.	Pitch diameter of thread, max.	Gage
202.	Depth of spiral groove	Gage
203.	Total length max.	Gage
204.	Length of relief of spiral groove max.	Gage
205.	Width of relief of spiral groove, max.	Measure
206.	Location of spiral groove	Measure
207.	Location of spiral groove relief	Measure
208.	Direction of turn of spiral groove, incorrect	Visual
209.	Number of turns of spiral groove, incorrect	Visual
210.	Depth of relief of spiral groove min.	Gage

4.3.2.13 Tube, Outside, Dwg. D8593684, covering a detail of Dwg. F8593670.

CRITICAL

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MAJOR	
101. Pitch diameter of thread, min. (both ends)	Gage
102. Major diameter of thread, min. (both ends)	Gage
103. Concentricity of pitch diameter of thread with inside diameter at datum locations	Gage
104. Diameter, inside	Gage
105. Length of effective thread, min.	Gage
106. Thickness of wall, min.	Gage
107. Straightness	Gage
108. Thread damaged or not full	Visual
109. Metal defective	Visual
110. Protective finish inadequate or defective	Visual
111. Burr	Visual
112. Foreign matter	Visual
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of body, max.	Gage
203. Total length, max.	Gage
204. Length of thread, max.	Gage

4.3.2.14 Cap Dwg. 8593685-77 covering a detail of Dwg. F8593670.

CRITICAL	
None defined	
MAJOR	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of mounting holes	Gage
104. Symmetry of length between lugs with thread	Gage
105. Symmetry of outerside of lugs with innerside of lugs	Gage
106. True position of mounting holes with thread	Gage
107. True position of lugs with thread	Gage
108. Diameter of cavity at datum location, max.	Gage
109. Thickness of base, min.	Gage
110. Pellet missing	Visual
111. Depth of pellet hole (max.)	Gage
112. Protective finish inadequate or defective	Visual
113. Burr	Visual
114. Foreign matter	Visual
115. Thread damaged or not full	Gage
116. Metal defective	Visual
117. Burr or sharp corner	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter at datum location, min.	Gage
203. Diameter of mounting hole	Gage
204. Length from outerside to outerside of lugs	Gage
205. Length between lugs	Gage
206. Location of mounting hole	Gage

4.3.2.15 Ring, Compression, Dwg. C8594741, covering a detail of Dwg. F8593670.

CRITICAL	
None defined	
MAJOR	
101. Diameter at basic internal location, max.	Gage
102. Diameter of cavity, max.	Gage
103. Metal defective	Visual
104. Protective finish, inadequate or defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter at basic internal location, min.	Gage
202. Diameter of cavity, min.	Gage

4.3.2.16 Metal Parts Assembly, Dwg. F8593670.

CRITICAL	
None defined	
MAJOR	
101. Any part missing except as otherwise classified	Visual
102. Improper assembly	Visual
103. Thread damaged or not full	Visual
104. Marking missing, incorrect or illegible	Visual
105. Burr	Visual
106. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 20261O-22T06:30Z
Check the source to verify that this is the currentversion before use.

Defect	Method of Inspection
MINOR	
201. Shipping plug missing	Visual
202. Tube part number not adjacent to tube stop ring	Visual
203. Varnish, missing or inadequate	Visual
204. Protective finish inadequate or defective	Visual
4.3.2.17 Packing in Fiber Container, Dwg. F8797383	
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual
4.3.2.18 Packing in Wooden Box, Dwg F8797384.	
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware, missing, loose, broken or improperly engaged	Visual/Manual
203. Strapping missing, loose, broken or improperly assembled	Visual/Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:30Z
Check the source to verify that this is the current version before use.

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.- Fifty units from each heat treated lot of components shall be tested to determine compliance with the requirements of the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.2 Hydrostatic pressure.- Each outside, inside and telescoping tube shall be subjected to an internal hydrostatic pressure. Failure of any tube to comply with the requirements of 3.5 shall be cause for rejection of the tube.

4.3.3.3 Shear pin strength.- Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with the requirements of 3.6 shall be cause for rejection of the lot.

4.3.3.4 Structural strength.- A sample of 15 complete assemblies shall be selected and subjected to the structural strength test. Failure of any assembly to comply with the requirements of 3.7 shall be cause for rejection of the lot.

4.3.3.5 Breakaway torque.- A breakaway torque shall be applied to each assembly threaded joint as listed in Table II. Failure of any assembly to comply with the requirements of column C of Table II shall be cause for rejection of the assembly.

4.3.3.6 Protective finish.- The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable military standard listed on the drawing.

4.3.3.7 Residual magnetism.- Each metal parts assembly shall be subjected to a test for excessive residual magnetism. Failure of any assembly to comply with the requirements of 3.10 shall be cause for rejection of the assembly.

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8593670.

4.4 Test methods and procedures.

4.4.1 Hardness.- Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the hardness requirements specified on the detail drawings.

4.4.2 Hydrostatic pressure.- Prior to application of protective finish, the outside, inside and telescoping tube shall be mounted, individually in an approved device and subjected to an internal hydrostatic pressure in accordance with the requirements of 3.5. The testing device used shall internally

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:30Z
Check the source to verify that this is the current version before use.

4.4.2 (Cont.)

seal the ends of the tubes within 1½ inches from each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 Shear pin strength.- After application of protective finish, the pin shall be subjected to the shear strength test in accordance with the requirements of 3.6. The test shall be performed in the specified fixture in conjunction with a tensile testing machine.

4.4.4 Structural strength.- The catapult shall be placed in a tensile testing machine and subjected to the specified tension and compression loads as specified in 3.7. The test shall be performed as follows:

a. Apply a preload of 500 pounds in tension and adjust the dial indicator to zero inches.

b. Apply the load in tension so as to move the mounting points away from each other.

c. Release the load in tension to 500 pounds and record the dial indicator reading.

Repeat steps "b" and "c"

d. Apply a preload of 500 pounds in compression and adjust the dial indicator to zero inches.

e. Apply the load in compression so as to move the mounting points toward each other.

f. Release the compression load to 500 pounds and record the dial indicator reading.

Repeat steps "e" and "f"

4.4.5 Breakaway torque.- A torque wrench set at the torque requirements specified in 3.8 shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of torque.

4.4.6 Protective finish.- Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.4.7 Residual magnetism.- The catapult shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the catapult 6 inches from and at the same height as the compass, in a north-south horizontal position (block end at south) with the compass set in an east-west heading as shown in figure 1. The catapult shall be moved slowly

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

4.4.7 (Cont.)

past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A. Packing and marking shall be in accordance with drawings F8797383 and F8797384.

5.2 Data cards.- Data card information shall be as specified in MIL-A-2550.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

Custodian:

Army - MU

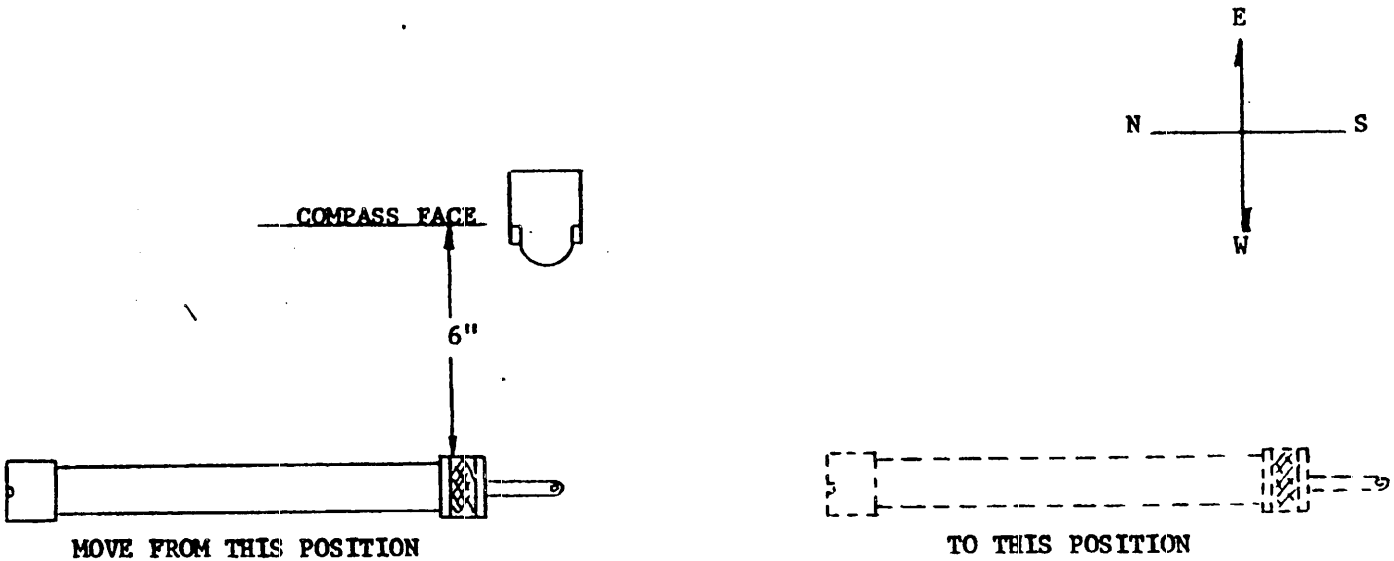
Preparing Activity:

Army - MU

Project No. 1375-A823

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

PILOT'S STANDBY COMPASS, TYPE AN-5766



PLAN VIEW

NOTE:

CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

FIGURE 1

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN CATAPULT, AIRCRAFT EJECTION SEAT, M4A1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.

STRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in mitting comments and suggestions for improvements. All users of military standardization documents are invited to provide gestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and iled. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the oblems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an knowledgement will be mailed to you within 30 days to let you know that your comments were received and are being naidered.

OTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of ecification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

EPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
ENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commanding Officer
Frankford Arsenal
Attn: SMUFA-N1100
Philadelphia, PA 19137

Source: https://assist.dla.mil -- Downloaded: 202640-22T06:30Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Inc Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:30Z
Check the source to verify that this is the current version before use.